SUPPLEMENTARY MATERIALS FOR "SPREAD TOO THIN: THE IMPACT OF LEAN INVENTORIES"

Appendix A Empirics

This section includes further details on the JIT adopters sample, an alternative measure of JIT, and a series of additional empirica exercises which supplement the results in the main text.

A.1 Sample construction

My data come from three sources. First, I use annual Compustat data to obtain information on firm-level inventory holdings, sales, and other outcomes. Second, I use the GSCPI in the regressions that estimate the JIT sensitivity to aggregate supply conditions. Lastly, for the weather regressions reported in Appendix A.3, I collect county-level weather event data from NOAA and map weather events to firm headquarter zip codes.

My Compustat sample covers the years 1971-2019. I keep only manufacturers (two-digit NAICS codes 31, 32, and 33). In addition, I drop firm-years in which acquisitions exceed 5% of total assets to avoid the influence of large mergers. To guard against measurement error, I keep only those firms with non-missing and positive book value of assets, number of employees, inventories, and sales. All variables are winsorized at the top and bottom 0.5% of the empirical distribution.

As discussed in the text, I define input inventories as the sum of raw material and work-in-process (`invrm+invwip`). This empirical definition also accords with the structural model developed in the main text in which producers carry stocks of inputs across time. My final sample consists of 5,912 unique firms. Table A1 reports summary statistics for the variables used.

A.2 Validating inventory-to-sales-based measure of JIT

In this section, I explore the robustness of my measure of JIT to alternative assumptions. I then explore an alternative measure of JIT that identifies the adoption of lean production based on fi-

Table A1: Compustat Summary Statistics

Variable name	Compustat code definition	Mean	Median	Standard deviation	25%	75%
Earnings growth	$\frac{\Delta \texttt{ib}}{\texttt{ib}_{-1}}$	-0.285	-0.069	6.083	-0.805	0.449
Employment growth	$\frac{\Delta \texttt{emp}_t}{\texttt{emp}_{t-1}}$	-0.001	0.000	0.294	-0.083	0.100
IHS earnings	$ihs(\texttt{ib}_t)$	0.839	0.850	3.231	-1.578	3.201
Log sales	$\log(\texttt{sale}_t)$	4.381	4.373	2.338	2.824	5.977
Sales growth	$\frac{\Delta \texttt{sale}_t}{\texttt{sale}_{t-1}}$	0.067	0.055	0.413	-0.062	0.178
Log sales per worker	$\frac{\texttt{sale}_t}{\texttt{emp}_t}$	4.978	4.982	0.868	4.454	5.516
JIT adoption		0.676	1.000	0.468	0.000	1.000

Note: The table reports summary statistics for the relevant variables in Section 2 of the text. The sample is constructed from Compustat Fundamentals Annual files for 1980-2019. Sample consists of 5,912 unique firms.

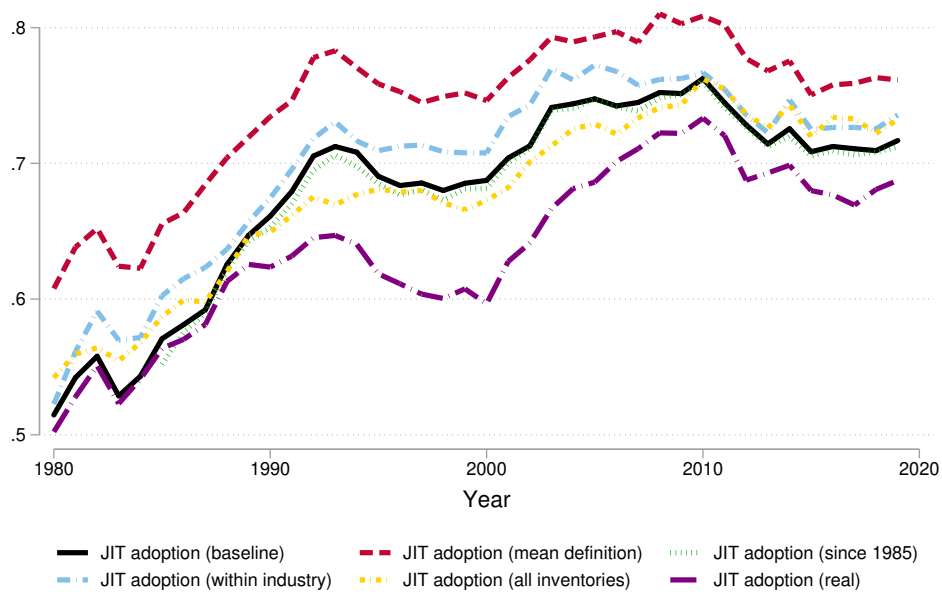
nancial news reports, press releases, and Form 10-K filings. These data were kindly provided to me by William Wempe, from his joint work with Michael Kinney, and Xiaodan Gao. See Kinney and Wempe (2002) and Gao (2018) for further details. This measure identifies JIT adoption for 177 manufacturing firms in my Compustat sample.

A.2.1 Measuring JIT under alternative assumptions

In this section, I consider alternative assumptions in the construction of my JIT measure. First, I define JIT based on a comparison of inventory-to-sales ratios relative to pre-1980s mean inventory-to-sales ratios within sectors rather than median inventory-to-sales ratios. Second, I measure JIT by comparing inventory-to-sales to historical inventory-to-sales within a sector from 1971-1985 rather than 1971-1980. Third, I define JIT by comparing inventory-to-sales to historical inventory-to-sales within a narrower 4-digit NAICS industry rather than 2-digit NAICS sector. Fourth, I define JIT based on total inventories (raw material, work-in-process, and finished goods) rather than only input inventories (raw material and work-in-process). Fifth, I define JIT based on real inventory-to-sales ratios rather than nominal inventory-to-sales ratios, using manufacturing inventory and sales deflators from the National Income and Product Accounts (NIPA) tables. Figure A1 plots the evolution of the share of measured JIT adopters in my sample according to the baseline definition

as well as these alternative definitions. Overall, these various measures of JIT evolve similarly over time, with my baseline definition generally residing in the middle of the range of these various measures.

Figure A1: Alternative Measures of JIT Adoption



Note: The figure plots alternative measures of the share of JIT adopters in the sample based on variations to the definition in equation (1) of the main text.

A.2.2 A text-based measure of JIT

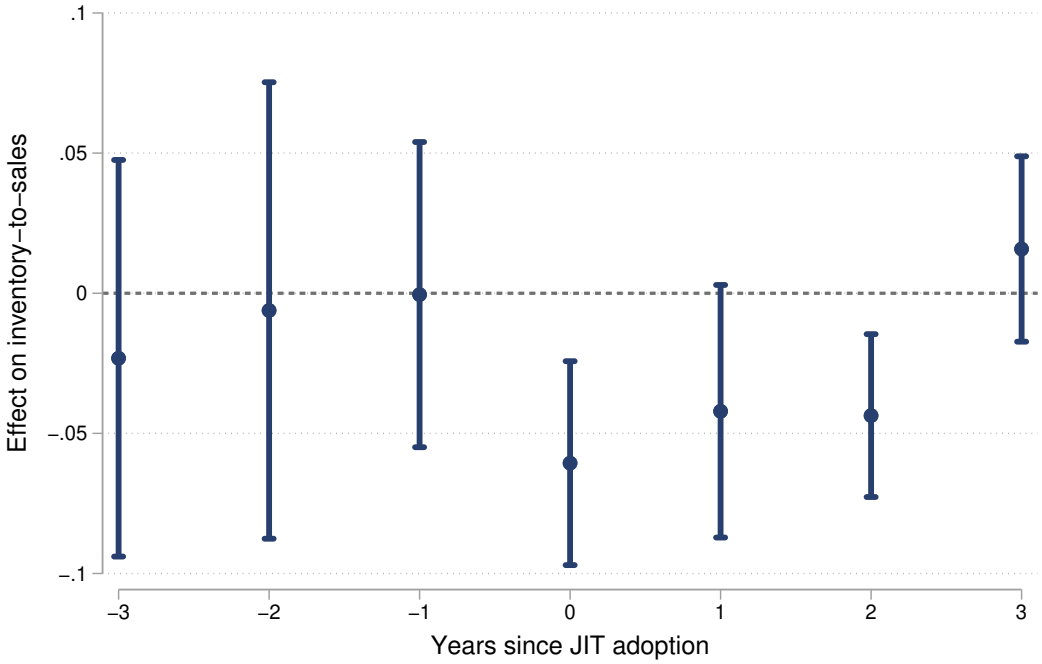
I next examine a measure of JIT that identifies adoption of lean production based on financial news reports, press releases, and Form 10-K filings for a narrower set of manufacturing firms.

I start by regressing inventory-to-sales ratios on this alternative measure of JIT adoption to verify that inventory-to-sales ratios decline in the years following adoption for these firms. I estimate the following regression,

$$y_{ijt+h} = \gamma \text{adopt}_{ijt} + \delta_{jt} + \delta_i + \varepsilon_{ijt},$$

where the outcome of interest is the inventory-to-sales ratio, and adopt_{ijt} is an indicator taking on a value of one only in the recorded year of adoption. Industry-by-year and firm fixed effects are

Figure A2: Validation of JIT Indicator



Note: The figure plots the estimated effect of JIT adoption on the level of inventory-to-sales. 95% confidence bands are displayed alongside point estimates.

specified. The figure plots 95% confidence intervals for a three-year window around the recorded date of adoption and shows that inventory holdings decline in the year of adoption and over the subsequent two years.

Next, I compare the two measures of JIT adoption. Table A2 reports the share of JIT firms in a given 3-digit NAICS industry (and identified through the text-based approach). Based on these reported frequencies, we see that most of the 177 JIT firms identified through the text-based approach reside in NAICS sector 33 and, specifically, within fabricated metal manufacturing, machinery manufacturing, computer and electronic product manufacturing, electrical equipment, appliance, and component manufacturing, and transportation equipment manufacturing. The final column of Table A2 reports each industry's average inventory-to-sales ratio from 1980 to 2019 relative to its pre-1980 median. Across all industries, we see a decline in the inventory-to-sales ratio relative to pre-1980, albeit to varying degrees.

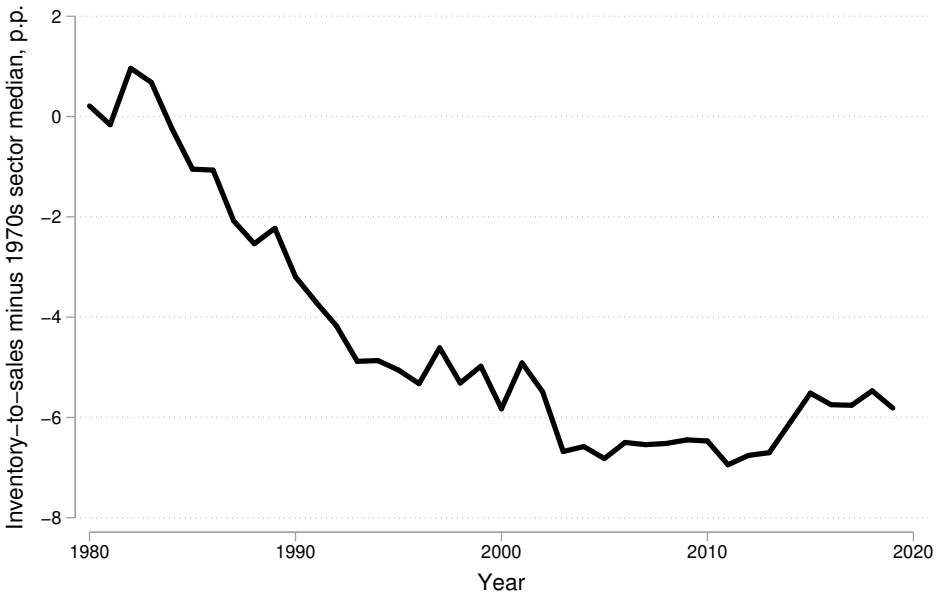
Table A2: Text-Based JIT Adopters by Industry

Description	3-digit NAICS	Percent of text-based adopters	Inventory-to-sales relative to pre-1980s medians (percent)
Food manufacturing	311	1.96	-6.76
Textile mills	313	1.31	-0.05
Textile product mills	314	0.65	-0.78
Leather and allied product manufacturing	316	1.31	-4.93
Wood product manufacturing	321	0.65	-0.74
Paper manufacturing	322	1.31	-2.97
Printing and related support activities	323	0.65	-5.61
Chemical manufacturing	325	2.61	-1.74
Plastics and rubber manufacturing	326	1.96	-4.05
Nonmetallic mineral product manufacturing	327	0.65	-2.73
Primary metal manufacturing	331	3.27	-0.03
Fabricated metal product manufacturing	332	7.19	-7.15
Machinery manufacturing	333	15.03	-6.42
Computer and electronic product manufacturing	334	35.29	-5.36
Electrical equipment, appliance, and component manufacturing	335	7.19	-5.19
Transportation equipment manufacturing	336	11.76	-5.78
Furniture and related product manufacturing	337	3.27	-9.39
Miscellaneous manufacturing	339	3.92	-3.34

Note: The table lists 3-digit NAICS industry codes and descriptions as well as the share of JIT of firms identified through the text-based approach. The final column reports the average inventory-to-sales ratio in each industry relative to the pre-1980 median.

Figure A3 pools across the text-based JIT adopters and plots their average inventory-to-sales ratio, in each year, relative to their respective pre-1980s sector medians. The figure indicates that on average inventory-to-sales ratios among text-based JIT firms declined relative to their respective pre-1980 median inventory-to-sales ratios.

Figure A3: Inventory-to-sales Among Text-Based JIT Adopters



Note: The figure plots inventory-to-sales ratio of text-based JIT adopters relative to their respective 1970s sector median inventory-to-sales ratios.

A.3 JIT and weather events

In addition to being more sensitive to aggregate supply conditions, I find that JIT adopters appear to be more sensitive to weather events faced by their suppliers. I examine this by merging my data with county-level weather events from NOAA using the Compustat Segment Files and links from Barrot and Sauvagnat (2016). I then estimate the following regression:

$$y_{ist} = \psi_1 \text{JIT}_{it-1} + \psi_2 \text{WeatherEvent}_{st} + \psi_3 \big[\text{JIT}_{it-1} \times \text{WeatherEvent}_{st}\big] + \mathbf{X}'_{ist}\beta + \text{FE} + \omega_{ist}. \quad (1)$$

I consider two ways of defining the "WeatherEvent" regressor: (i) as an indicator for a weather event occurring in the zip code where supplier s is headquartered in a given year and (ii) as the dollar value of property damage caused by the weather event. I collect information on county-level weather events from NOAA and link these events to public firm headquarter zip codes via the aforementioned Barrot and Sauvagnat (2016) links.

Ideally, one would want to link upstream weather events to the zip codes in which suppliers' production takes place. Compustat data is limited in this respect since one cannot necessarily assume that production occurs at or near a firm's headquarters. Nonetheless, weather events may disrupt other relevant operations which might take place at a firm's headquarters such as logistics. Overall, I interpret this data limitation as a form of measurement error which likely biases my estimates toward zero.

Table A3 provides four sets of results which summarize the estimated sales response to supplier weather events. The first two columns report the effect of a weather event on sales when specifying the weather event indicator variable. The final two columns instead report the property damage with respect to weather events. The point estimates on the interaction between a supplier weather event and a JIT customer are negative across all specifications and are statistically significant when controlling for supplier-by-year fixed effects. The latter estimates, reported in columns (2) and (4), control for upstream time-variation which includes year-specific supplier characteristics such as age and size, as well as other unobserved shocks that suppliers face in a given year. Given the

Table A3: JIT Adoption and Weather Events

	(1)	(2)	(3)	(4)
Weather event indicator	-0.003			
	(0.036)			
Weather event indicator × JIT	-0.007	-0.195***		
	(0.038)	(0.069)		
Property damage			-0.0002	
			(0.002)	
Property damage × JIT			-0.0003	-0.010***
			(0.002)	(0.003)
Fixed effects	Firm, Supplier, Year	Firm, Supplier×Year	Firm, Supplier, Year	Firm, Supplier×Year
Firms	196	68	196	68
Observations	1885	317	1885	317

Note: The table reports panel regression results based on regression (1). The dependent variable is log sales. The control variables specified include lagged JIT indicator, lagged log capital stock, and firm age in sample. Standard errors are double clustered at the customer-supplier level. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

more robust set of controls specified in these regressions, columns (2) and (4) reflect my preferred specifications.

Through the series of links required to estimate these regressions, the sample size is reduced considerably.[1] Nonetheless, in my preferred specifications, I find that, on average, supplier weather events in my sample predict an additional 19.5% decline in JIT firm sales. Furthermore, a 1% increase in the property damage caused by a given weather event is associated with a 1% excess sales contraction among JIT firms relative to non-JIT firms.

[1]Building this sample requires linking weather events to firm (supplier) headquarters in Compustat, then linking these suppliers to their customers (through the Segment files), and finally linking the customers to their JIT adoption status.

A.4 The effects of switching out of JIT

While the regressions that I estimate are linear and imply symmetric effects between JIT adopters and non-adopters, it is useful to examine the outcomes specifically for firms that switch out of JIT adoption. I construct an indicator that is equal to one when a firm switches out of JIT. I then re-estimate regressions (2) and (3) to determine whether switching out of JIT is associated with a decrease in sales and an increase in firm-level volatility. I find that this is indeed the case. Tables A4 and A7 report the results when using an indicator for switching out of JIT. These results indicate that the estimated effect of JIT adoption on sales, sales per worker, and earnings is stronger than JIT switching, but are quantitatively symmetric for the firm volatility results.

Table A4: JIT Switching and Firm Performance

	Sales	Sales per worker	Earnings
JIT switch	-0.149***	-0.121***	-0.167***
	(0.017)	(0.015)	(0.052)
Fixed effects	Firm, Sector × Year	Firm, Sector × Year	Firm, Sector × Year
Firms	5,427	5,427	5,427
Observations	55,592	55,592	55,592

Note: The table reports panel regression results based on regression (2) in the main text, but with an indicator for switching out of JIT rather than for adopting JIT. The dependent variables are log sales, log sales per worker, and the inverse hyperbolic sine of earnings. Earnings are defined as income before extraordinary items. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A5: JIT Switching and Firm Volatility

	Sales growth volatility	Employment growth volatility	Earnings growth volatility
JIT switch	0.134***	0.058**	0.112*
	(0.035)	(0.024)	(0.063)
Fixed effects	Sector × Year	Sector × Year	Sector × Year
Firms	2,800	2,796	2,800
Observations	18,975	18,903	18,955

Note: The table reports panel regression results based on regression (3) in the main text, but with an indicator for switching out of JIT rather than adopting JIT. The dependent variables are rolling five-year standard deviations of firm sales growth, employment growth, and earnings growth. Lagged log of total assets and age in sample are specified as controls. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

A.5 Additional evidence consistent with model mechanism

In this section, I document additional evidence that is consistent with the mechanism governing JIT adoption in the model.

A.5.1 Lower mean and variance of fixed ordering costs among JIT producers

The lower mean fixed ordering cost assumption is directly linked to the lower inventory-to-sales ratio observed after JIT adoption. This is generated by the (s,S) ordering decision in the model. At an extreme, if fixed ordering costs were zero, then no firm holds inventories in the model. Table A6 reports the average response of the inventory-to-sales ratio in my sample following JIT adoption.

Table A6: JIT and inventory-to-sales

	Inventory-to-sales
JIT	-0.876***
	(0.021)
Fixed effects	Sector × Year
Firms	5,427
Observations	55,592

Note: The table reports panel regression results based on regression (2) in the main text. Lagged log of total assets and age in sample are specified as controls. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

The lower variance of fixed order costs is consistent with lower firm sales and earnings volatility. In addition, this is also consistent with lower inventory investment volatility which I show in the table below.

A.5.2 Sunk and fixed adoption costs

The sunk cost-continuation cost assumption is made in order to generate persistence in the JIT adoption decision. Table A8 reports empirical JIT transition probabilities, which imply that exit from JIT is rare once adoption occurs, consistent with the presence of sunk adoption costs.

Table A7: JIT and inventory investment volatility

	Inventory growth volatility
JIT	-0.496***
	(0.030)
Fixed effects	Sector × Year
Firms	2,800
Observations	18,975

Note: The table reports panel regression results based on regression (3) in the main text. Lagged log of total assets and age in sample are specified as controls. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A8: JIT transition probabilities

	$JIT_{t+1} = 0$	$JIT_{t+1} = 1$
$JIT_t = 0$	0.856	0.144
$JIT_t = 1$	0.056	0.944

Note: The table reports empirical JIT transition probabilities from 1980 to 2019 based on the baseline JIT measure.

In addition, based on a simple linear probability model, lagged JIT adoption predicts future JIT adoption. The results are reported in Table A9. I include firm fixed effects in this regression to ensure that the results reflect persistence and not permanent heterogeneity.

Table A9: Persistence in JIT adoption decision

	JIT_t
JIT_{t-1}	0.529***
	(0.010)
Controls	Yes
Fixed effects	Firm, Sector × Year
Firms	4,724
Observations	43,782

Note: The table reports panel regression results based on regression (2) in the main text. Total assets, age in sample, and log sales are specified as controls. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

A.5.3 Persistence in JIT firm outcomes

The assumptions made about the fixed ordering cost distribution faced by JIT firms as well as the endogenous persistence in JIT adoption assumed by the sunk-continuation costs jointly imply greater persistence of firm-level outcomes such as inventory-to-sales ratios. Table A10 provides evidence consistent with this. According to the results in the table, JIT adoption is associated with more persistent inventory-to-sales ratios, consistent with smoother and more continuous inventory adjustment.

A.5.4 Redirecting resources to production following JIT adoption

In my model, the positive correlation between JIT and firm sales is partly due to positive selection into JIT. In addition, the model implies that JIT itself also generates higher sales because firms incur fewer costs associated with order fulfillment and the costly storing of idle stocks of inventory.

Table A10: JIT and inventory-to-sales persistence

	Inventory-to-sales$_t$
Inventory-to-sales$_{t-1}$	0.590***
	(0.013)
JIT$_t$	-0.107***
	(0.038)
JIT$_t$ $\times$ Inventory-to-sales$_{t-1}$	0.187***
	(0.017)
Controls	Yes
Fixed effects	Firm, Sector $\times$ Year
Firms	5,326
Observations	44,384

Note: The table reports panel regression results based on regression (2) in the main text. Total assets, age in sample, and log sales are specified as controls. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

JIT firms are able to redirect these resources to production. Table A11 provides further evidence consistent with this mechanism.

Table A11: Additional Results on JIT and Firm Performance

	Employment	Capital expenditures	Cash-to-assets	Order backlog
JIT	0.023**	0.036***	0.183***	-0.041**
	(0.010)	(0.012)	(0.028)	(0.019)
Fixed effects	Firm, Sector $\times$ Year	Firm, Sector $\times$ Year	Firm, Sector $\times$ Year	Firm, Sector $\times$ Year
Firms	5,427	5,417	5,200	2,434
Observations	55,592	55,249	51,307	24,747

Note: The table reports panel regression results based on regression (2) in the main text. The dependent variables are log employment, the inverse hyperbolic sine of capital expenditures, log cash-to-total assets, and log order backlog. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

In the data, JIT adoption is associated with about a 2.3% increase in employment and a 3.6% increase in capital expenditures. These two results are consistent with JIT adopters directing resources toward the use of inputs in the production process. In addition, JIT firms tend to hold more

cash relative to total assets, suggesting they have more resources available to possibly direct towards production rather than keeping cash tied up in relatively larger inventory stocks (Gao, 2018). Furthermore, JIT firms tend to experience fewer order backlogs, suggesting that they might transform inputs into final goods for sale more rapidly or efficiently than non-JIT producers.

My model abstracts away from capital investment, cash holdings, and order backlogs among the heterogeneous final goods producers. However, JIT allows these firms to hire more labor in the model and make greater use of their material inputs to produce, thus leading to higher sales.

Regarding lower unconditional firm volatility, JIT can be interpreted as an investment in more reliable delivery of inputs, on average. In the model, JIT producers face a fixed order cost distribution that is characterized by a lower mean and variance. The latter yields a reduction in micro-level ordering uncertainty which enables JIT firms to smooth their ordering cycles. This translates to smoother material and labor input usage, and smoother sales. Though I assume that JIT and non-JIT producers face different fixed ordering cost distributions, the estimation of the model parameters suggests that JIT firms indeed face less dispersion in fixed ordering costs which can be interpreted as less uncertainty associated with delivery delays.

Another possible mechanism which could link JIT with higher sales and less firm-level volatility is that JIT leads firms to improve their own demand forecasts through the adoption of new technologies. This mechanism, which is not featured in my model, could allow firms to generate greater sales. At the same time, more accurate forecasts could translate to smoother firm-level outcomes.

A.6 Partial observability of JIT adoption

My baseline measure of JIT measures JIT adoption from 1980 onward. A limitation of this approach is that my measure implies that many firms suddenly adopted JIT in 1980. As a result, my measure may not detect firms that adopted JIT prior to 1980, particularly in industries that transitioned early. If this is the case, then I may be only detecting later adopters while erroneously classifying earlier adopters as non-adopters.

If my measure erroneously classifies early adopters as non-adopters, then this should attenuate

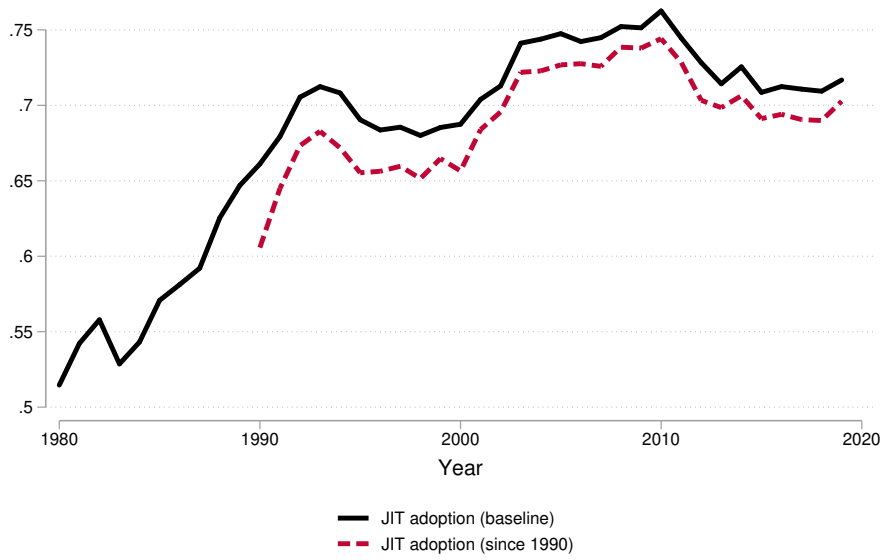
the estimated differences between JIT and non-JIT firms. However, it could also be the case that JIT happened sooner for certain types of firms (e.g., large and productive firms). If so, then my empirical estimates could be biased in either direction due to selection. To determine how serious of an issue this may be for my empirical results, I conduct a couple of robustness exercises.

A.6.1 Alternative benchmark year for JIT measurement

First, I construct another version of my measure of JIT, but benchmarked to 1990 (i.e., comparing firms' inventory-to-sales to their historical pre-1990 industry's median inventory-to-sales).

The figure below plots what the share of JIT adopters is when I use the 1990 benchmark year rather than 1980 benchmark year. Consistent with increased misclassification of early adopters, the later benchmark detects a smaller fraction of adopters.

Figure A4: Share of JIT adopters, varying benchmark year used for measurement



Note: The figure plots the measured share of JIT adopters in my sample based on different benchmark years.

With this additional measure of JIT in hand, I re-estimate my regressions using only data from 1990 to 2019, and I compare the results across the two measures. Tables A12, A13, and A14 report the results. The results using the 1990-based measure of JIT are similar to those that use the baseline

1980-based measure, however, the results using the 1990-based measure tend to yield estimates that are smaller in magnitude, consistent with the misclassification of early adopters leading to attenuation of the estimates.

Table A12: JIT Adoption and Firm Performance

	Sales		Sales per worker		Earnings	
	1980	1990	1980	1990	1980	1990
JIT	0.338***	0.316***	0.311***	0.303***	0.647***	0.636***
	(0.018)	(0.018)	(0.015)	(0.015)	(0.065)	(0.065)
Firms	4,500	4,500	4,500	4,500	4,500	4,500
Observations	41,233	41,233	41,233	41,233	41,233	41,233

Note: The table reports panel regression results based on regression (2) of the text. The dependent variables are log sales, log sales per worker, and the inverse hyperbolic sine of earnings. Earnings are defined as income before extraordinary items. The control variables specified are firm age in the sample and log total assets. Firm and sector-by-year time fixed effects are specified, where sectors are defined at the two-digit NAICS level. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A13: JIT Adoption and Firm Volatility

	Sales growth vol.		Emp. growth vol.		Earnings growth vol.	
	1980	1990	1980	1990	1980	1990
JIT	-0.248***	-0.234***	-0.049	-0.041	-0.046	-0.059
	(0.032)	(0.030)	(0.030)	(0.029)	(0.054)	(0.052)
Firms	2,149	2,149	2,147	2,147	2,149	2,149
Observations	12,855	12,855	12,809	12,809	12,841	12,841

Note: The table reports panel regression results based on regression (3) of the text. The dependent variables are rolling five-year standard deviations of firm sales growth, employment growth, and earnings growth. Lagged log of total assets and age in sample are specified as controls. Sector-by-year fixed effects are also specified where sectors are defined at the two-digit NAICS level. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A14: JIT Adoption and Supply Chain Pressures

	Sales 1980	Sales 1990	Sales 1980	Sales 1990
Supply chain pressure	0.012	0.009		
	(0.012)	(0.012)		
Supply chain pressure × JIT	-0.023**	-0.020**	-0.022**	-0.019*
	(0.009)	(0.009)	(0.009)	(0.009)
Fixed effects	Firm	Firm, Sector × Year	Firm	Firm, Sector × Year
Firms	2,766	2,766	2,766	2,766
Observations	20,608	20,608	20,608	20,608

Note: The table reports panel regression results from regression (4) of the text. The dependent variable is the log of firm sales. Lagged log of total assets, employment, and finished goods are specified as controls as well as firm age and contemporaneous unemployment rate, real GDP growth, and manufacturing PPI inflation. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

A.6.2 Dropping industry with low pre-1980 inventory dispersion

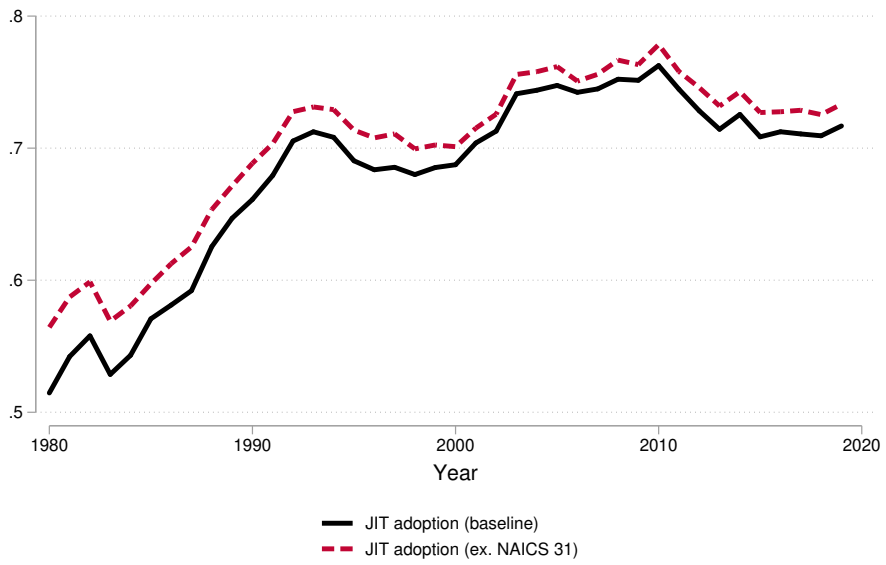
Second, I reason that if most firms in an industry are early adopters of JIT, then this industry should exhibit a relatively lower cross-sectional dispersion of inventories across firms within the industry. In this robustness exercise, I construct a new version of my baseline measure, dropping the sector with the lowest cross-sectional dispersion of inventories (NAICS 31).

There are some caveats to note about this exercise. First, if Compustat data happened to be noisier in the 1970s, then I may simply be dropping less noisy observations rather than early JIT adopters. Second, this exercise is primarily helpful if we think that an entire industry adopted JIT before other industries.

Figure A5 plots the evolution of JIT adoption with and without NAICS 31. When excluding NAICS 31, I estimate a slightly higher share of JIT adopters from 1980 onward, consistent with the NAICS 31 sector possibly reflecting early adopters.

I then re-estimate the empirical results in Tables A15, A16, and A17. Overall, I find that the estimates are similar across both measures of JIT and generally slightly larger in magnitude when excluding NAICS 31.

Figure A5: Share of JIT adopters, with and without NAICS 31



Note: The figure plots the baseline measure of JIT adoption as well as a version of the measure of JIT adoption that excludes NAICS 31.

Table A15: JIT Adoption and Firm Performance

	Sales		Sales per worker		Earnings	
	Baseline	ex. NAICS 31	Baseline	ex. NAICS 31	Baseline	ex. NAICS 31
JIT	0.305***	0.315***	0.279***	0.288***	0.546***	0.576***
	(0.015)	(0.016)	(0.014)	(0.014)	(0.050)	(0.051)
Firms	5,427	5,427	5,427	5,427	5,427	5,427
Observations	55,592	55,592	55,592	55,592	55,592	55,592

Note: The table reports panel regression results based on regression (2) of the text. The dependent variables are log sales, log sales per worker, and the inverse hyperbolic sine of earnings. Earnings are defined as income before extraordinary items. The control variables specified are firm age in the sample and log total assets. Firm and sector-by-year time fixed effects are specified, where sectors are defined at the two-digit NAICS level. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A16: JIT Adoption and Firm Volatility

	Sales growth vol.		Emp. growth vol.		Earnings growth vol.	
	Baseline	ex. NAICS 31	Baseline	ex. NAICS 31	Baseline	ex. NAICS 31
JIT	-0.190***	-0.196***	-0.047*	-0.058**	-0.116**	-0.112**
	(0.028)	(0.030)	(0.024)	(0.026)	(0.047)	(0.050)
Firms	2,800	2,800	2,796	2,796	2,800	2,800
Observations	18,975	18,975	18,903	18,903	18,955	18,955

Note: The table reports panel regression results based on regression (3) of the text. The dependent variables are rolling five-year standard deviations of firm sales growth, employment growth, and earnings growth. Lagged log of total assets and age in sample are specified as controls. Sector-by-year fixed effects are also specified where sectors are defined at the two-digit NAICS level. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

Table A17: JIT Adoption and Supply Chain Pressures

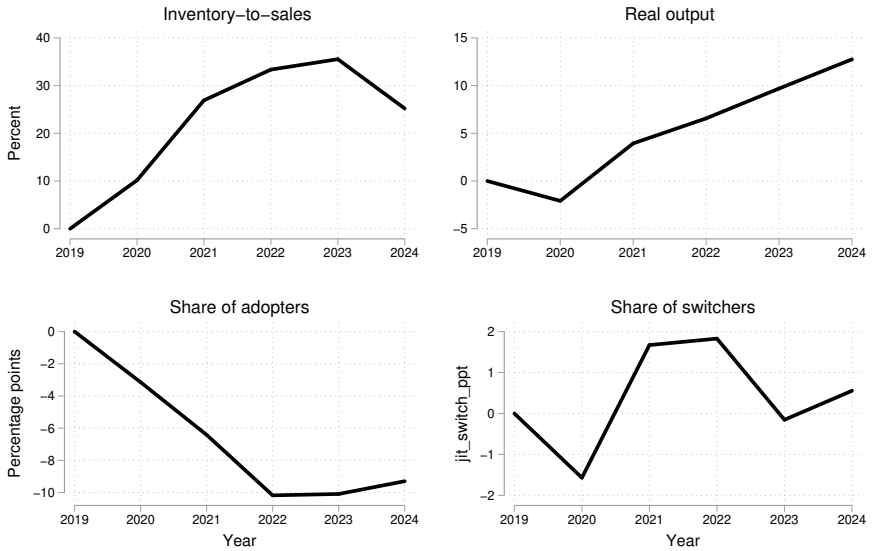
	Sales Baseline	Sales ex. NAICS 31	Sales Baseline	Sales ex. NAICS 31
Supply chain pressure	0.012	0.014		
	(0.012)	(0.013)		
Supply chain pressure × JIT	-0.023**	-0.026**	-0.022**	-0.022**
	(0.009)	(0.009)	(0.009)	(0.010)
Fixed effects	Firm	Firm, Sector × Year	Firm	Firm, Sector × Year
Firms	2,766	2,766	2,766	2,766
Observations	20,608	20,608	20,608	20,608

Note: The table reports panel regression results from regression (4) of the text. The dependent variable is the log of firm sales. Lagged log of total assets, employment, and finished goods are specified as controls as well as firm age and contemporaneous unemployment rate, real GDP growth, and manufacturing PPI inflation. Two-digit NAICS codes are specified in the sector-by-year fixed effects. Standard errors are double clustered at the firm and fiscal year levels. *** denotes 1% significance, ** denotes 5% significance, and * denotes 10% significance.

A.7 Post-2019 outcomes in the data

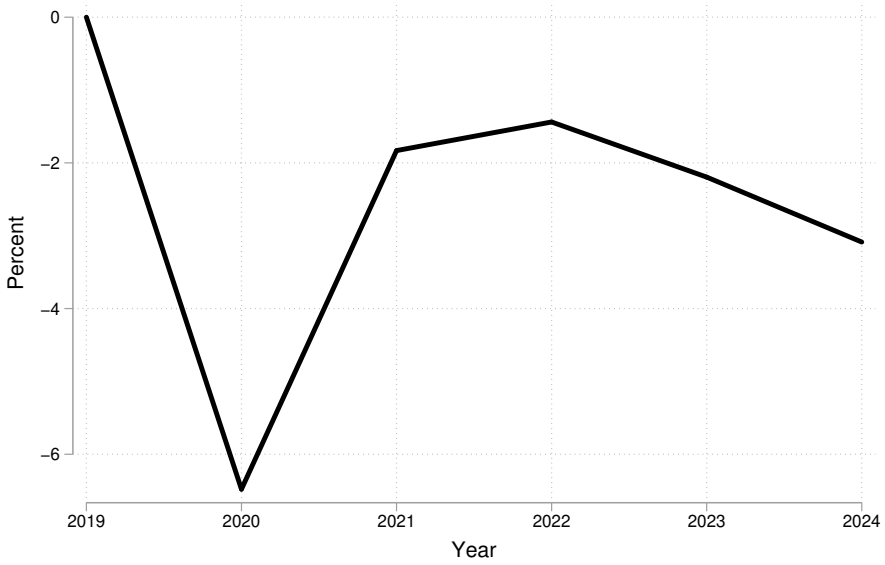
In this section, I compare the counterfactual transition paths in Figure 5 of the text with their empirical counterparts shown in Figures A6 and A7.

Figure A6: Post-2019 outcomes in the data



Note: The figure plots four panels, each depicting an annual time series since 2019 obtained from my sample. Inventory-to-sales, share of adopters, and share of switchers are all aggregated by averaging across firms in each year. Real output is defined as real GDP.

Figure A7: Post-2019 manufacturing industrial production



Note: The figure plots annual manufacturing industrial production since 2019.

Appendix B Model

In this section, I first provide details on how the final goods firm problem is reduced from a choice over inventory, s' and labor, n, to only a choice over inventory. Second, I report the expression for the price of the intermediate good, q. Finally, I detail the model's numerical solution.

B.1 Reducing the final goods firm problem to a choice over inventories

Final goods firms choose s' and n to maximize their value in the production stage. We can optimize out the static labor decision. The firm's payoff in the current period solves,

$$\max_{s' \in [0,s], n} zn^{\theta_n}(s - s')^{\theta_m} - c_m s'.$$

Solving for labor, n, we obtain,

$$n = \left[\frac{\theta_n}{w} z(s - s')^{\theta_m} \right]^{\frac{1}{1-\theta_n}}.$$

Substituting this into the maximization problem, we obtain

$$\max_{s'} (1 - \theta_n) \left[z \left(\frac{\theta_n}{w} \right)^{\theta_n} (s - s')^{\theta_m} \right]^{\frac{1}{1-\theta_n}} - c_m s',$$

or, equivalently,

$$\max_{s'} (1 - \theta_n) y(z, s, s') - c_m s'.$$

B.2 Intermediate goods price

The intermediate goods firm problem is:

$$\max_{K,L} qK^{\alpha} L^{1-\alpha} - RK - wL.$$

The solution to this problem yields the following closed-form expression for the price of the intermediate good,

$$q = \left(\frac{1+r}{\alpha}\right)^{\alpha}\left(\frac{w}{1-\alpha}\right)^{1-\alpha}.$$

B.3 Numerical solution

The model is solved using methods that are standard in the heterogeneous firms literature. The exogenous productivity process is discretized according to Tauchen (1986) which allows me to express the AR(1) process for log firm productivity as a Markov process. I select $N_z = 11$ grid points for idiosyncratic productivity and $N_s = 200$ grid points for the endogenous inventory state. Considering the binary adoption state, this implies that the discretized model has 4,400 grid points.

I can simplify the model by defining $p \equiv U_1(C, H) = \frac{1}{C}$ and reformulating the final goods firms' problem as the maximization of dividends weighted by marginal utility price, p. In doing so, firms weight current profits by p and discount expected future earnings by β (eliminating the time-varying discount factor, Λ_{t+1}). Defining the value function $v^A = pV^A$, the firm's reformulated problem is,

Stage 1: Adoption decision

$$v^A(z,s,a) = \max\left\{-pwc(a) + \int v^O(z,s,1,\xi)dF(\xi_A), \int v^O(z,s,0,\xi)dF(\xi_{NA})\right\}, \tag{2}$$

where

$$c(a) = \begin{cases} c_s & \text{if no JIT } (a=0) \\ c_f & \text{if JIT } (a=1), \end{cases}$$

Stage 2: Order decision

$$v^O(z,s,a,\xi) = \max\left\{-pw\xi + v^*(z,s,a), v^P(z,s,a)\right\}, \tag{3}$$

where the value of placing an order is

$$v^*(z,s,a) = \max_{s^* \geqslant s} \Big[-pq(s^* - s) + v^P(z,s^*,a) \Big], \tag{4}$$

and $v^P(z,s,a)$ is defined below. The firm's order problem delivers a threshold rule. In particular, a firm places an order if and only if the order cost draw is lower than a threshold order cost: $\xi < \xi^*(z,s,a)$ where

$$\xi^*(z,s,a) = \min\big(\max\big(\underline{\xi}, \widetilde{\xi}(z,s,a)\big), \overline{\xi}\big), \tag{5}$$

and

$$\widetilde{\xi}(z,s,a) = \frac{v^*(z,s,a) - v^P(z,s,a)}{pw}. \tag{6}$$

Stage 3: Production decision

$$\widetilde{s} = \begin{cases} s^*\big(z,s,a'(z,s,a)\big) & \text{if order placed} \\ s & \text{if no order placed.} \end{cases}$$

In the production stage, the firm selects labor, $n(z,\widetilde{s},s',a)$, and materials, $(\widetilde{s} - s')$, to maximize profits. Its value function in the production stage is:

$$v^P(z,\widetilde{s},a) = \max_{s' \in [0,\widetilde{s}]} p\Big[zn(z,\widetilde{s},s',a)^{\theta_n}(\widetilde{s}-s')^{\theta_m} - wn(z,\widetilde{s},s',a) - c_m s' \Big] + \beta \mathbb{E}\big[v^A(z',s',a')\big]. \tag{7}$$

I solve for the policy functions via value function iteration which is accelerated by the use of the MacQueen-Porteus error bounds (MacQueen, 1966; Porteus, 1971). This acceleration method makes use of the contraction mapping theorem to obtain bounds that produce a better update of the value function. The ergodic distribution of firms is obtained via nonstochastic simulation as in Young (2010).

I solve the model by initiating a guess for p, p_0. Using the household and intermediate goods producer's optimality conditions, I obtain the implied wage and intermediate goods price, w_0 and q_0, given the guess p_0. From here, I solve the firm's problem via value function iteration and

then obtain the ergodic distribution via nonstochastic simulation. Using the policies and ergodic distribution, I compute aggregates and the associated market clearing error from the household's optimality condition. I update the price based on this error using bisection.

For the unexpected shock exercises, I implement a standard shooting algorithm used to model deterministic dynamics. I fix the duration of the transition to a predetermined length T so that the model reaches steady state at $T+1$. I then solve the final goods firms' problem backwards, obtaining a set of time-indexed policy functions. Using these optimal decisions, I push the distribution of final goods firms forward beginning in the steady state. With the time-indexed policies and weights in hand, I compute aggregates at each point in time and iterate on prices until the final goods market clears in each period, $\frac{1}{p_t} = C_t$.

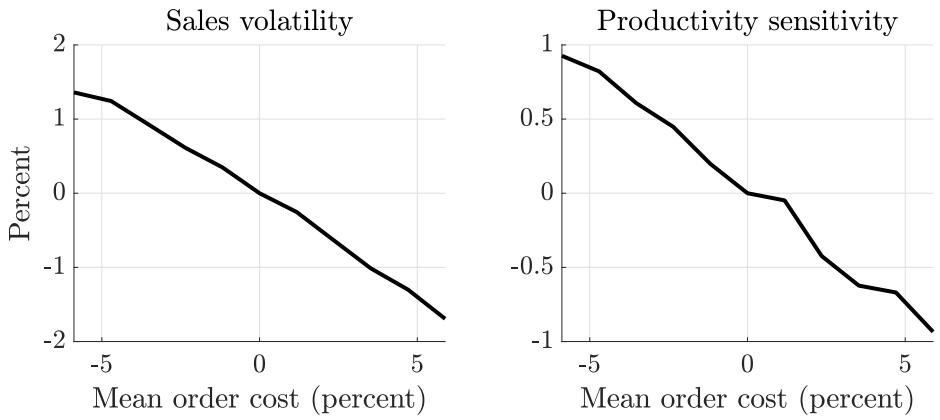
B.4 Lower volatility and higher sensitivity with JIT

JIT adopters face a fixed ordering cost distribution that has a lower mean and variance than the fixed ordering cost distribution of non-JIT firms. I find that, quantitatively, the lower variance of fixed ordering costs drives the reduction in sales volatility following adoption. Intuitively, the variance of the fixed ordering cost distribution affects the predictability of fixed ordering costs and therefore the timing of orders across and within firms. This in turn shapes the volatility of firm-level outcomes such as sales growth. Meanwhile, the lower mean of the fixed ordering cost distribution drives the sales sensitivity to idiosyncratic productivity result. Intuitively, the average fixed order cost affects firms' ordering inaction regions (i.e., the distance to the ordering threshold). As the average level of fixed order costs declines, inaction regions shrink which raises the probability of ordering, increasing the responsiveness of firms to productivity shocks.

I verify the intuition described above with a set of counterfactual exercises. I start by considering a simplified version of my baseline model without the JIT adoption decision. There is a distribution of firms that all face the same fixed ordering cost distribution. The fixed ordering cost distribution is calibrated to the same mean and variance as the fixed order cost distribution faced by non-JIT adopters in my baseline model. I then conduct a set of comparative static exercises.

First, I perturb the mean of the fixed ordering cost distribution in either direction, holding the variance fixed. The results are plotted in Figures B1. Based on this exercise, reducing the mean of the fixed ordering cost distribution leads to an increase in firms' sensitivity to idiosyncratic productivity, consistent with the intuition above. In addition, reducing the mean of the fixed ordering cost distribution raises firm sales volatility.

Figure B1: Perturbing mean order costs

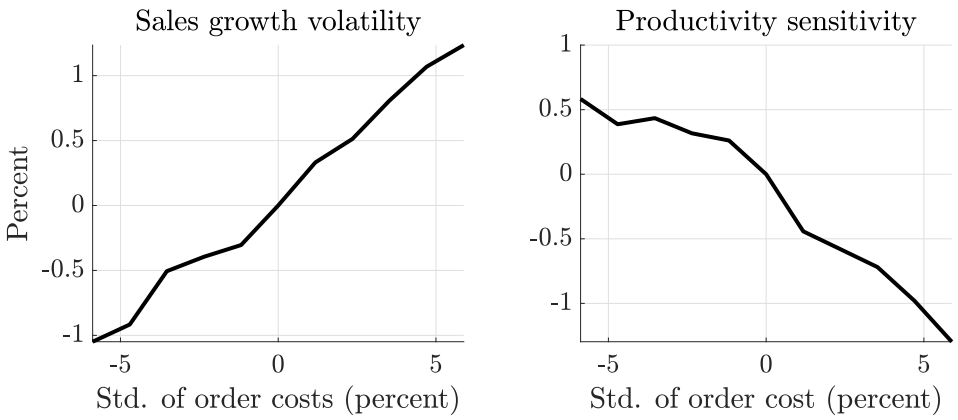


Note: The figure plots the percent change in firm sales volatility and sensitivity to idiosyncratic productivity for a given percent change in average fixed ordering costs (holding the variance fixed). Productivity sensitivity is defined as the coefficient from a regression of log firm sales on log firm productivity.

In a separate exercise, I instead perturb the variance of the fixed ordering cost distribution, holding the mean fixed. The results are plotted in Figure B2 and imply that reducing the variance of fixed ordering costs reduces sales volatility while increasing sensitivity of idiosyncratic productivity.

Taken together, a reduction in the mean and variance of fixed ordering costs, which characterizes the transition from JIT to non-JIT in the baseline model, will tend to unambiguously increase the sensitivity to idisoyncratic productivity while the effect on sales volatility is ambiguous. In my baseline JIT model, however, the reduction in the variance of fixed ordering costs faced by JIT firms when adopting JIT is much larger than the decline in average fixed ordering costs. So, quantitatively, the reduction in variance of fixed ordering costs is stronger, leading to a decline in sales volatility following adoption.

Figure B2: Perturbing variance of order costs



Note: The figure plots the percent change in firm sales volatility and sensitivity to idiosyncratic productivity for a given percent change in the variance of fixed ordering costs (holding the mean fixed). Productivity sensitivity is defined as the coefficient from a regression of log firm sales on log firm productivity.

Appendix C Estimation

In this section, I detail the estimation of the model and provide additional results relating to identification.

C.1 Simulated method of moments

The parameter vector to be estimated is $\theta = \big(\rho_z \;\; \sigma_z \;\; \overline{\xi}_{NA} \;\; \overline{\xi}_A \;\; c_s \;\; c_f \;\; c_m\big)'$. Estimating θ requires making a guess, θ_0, solving and simulating the model, and computing the different moments. I collect the targeted empirical moments in a stacked vector $m(X)$ which comes from my Compustat sample. I next stack the model-based moments, which depend on θ, in the vector $m(\theta)$. Finally, I search the parameter space to find the vector $\widehat{\theta}$ that minimizes the following objective,

$$\min_{\theta} \big(m(\theta) - m(X)\big)' W \big(m(\theta) - m(X)\big),$$

where W is the optimal weighting matrix, defined as the inverse of the covariance matrix of the moments. I compute the covariance matrix, clustering by firm following Hansen and Lee (2019). I estimate the parameter vector via particle swarm optimization, a standard stochastic global optimization solver.

The limiting distribution of the estimated parameter vector $\widehat{\theta}$ is,

$$\sqrt{N}(\widehat{\theta} - \theta) \xrightarrow{d} N(0, \Sigma),$$

where

$$\Sigma = \left(1 + \frac{1}{S}\right)\left[\left(\frac{\partial m(\theta)}{\partial \theta}\right)' W \left(\frac{\partial m(\theta)}{\partial \theta}\right)\right]^{-1},$$

and S is the ratio of the number of observations in the simulated data to the number of observations in the sample.[2] I obtain the standard errors of the parameters from Σ, where I compute the Jacobian, $\frac{\partial m(\theta)}{\partial \theta}$, via numerical differentiation.

[2] S is set to be approximately 6.5.

Given the lumpiness in ordering and inventories induced by the fixed ordering costs, the heterogeneous producers in my model are best characterized as establishments. Accordingly, when matching the model moments to the data, a simulated firm is assumed to consist of ten establishments. I classify a simulated firm as JIT according to my empirical definition of JIT, using the pre-1980 average inventory-to-sales ratio across all manufacturing sectors (approximately 0.13).

C.2 Identification

While the targeted moments jointly determine the parameters to be estimated, there are nonetheless moments that are especially important for pinning down certain parameters. I discuss their informativeness in turn.

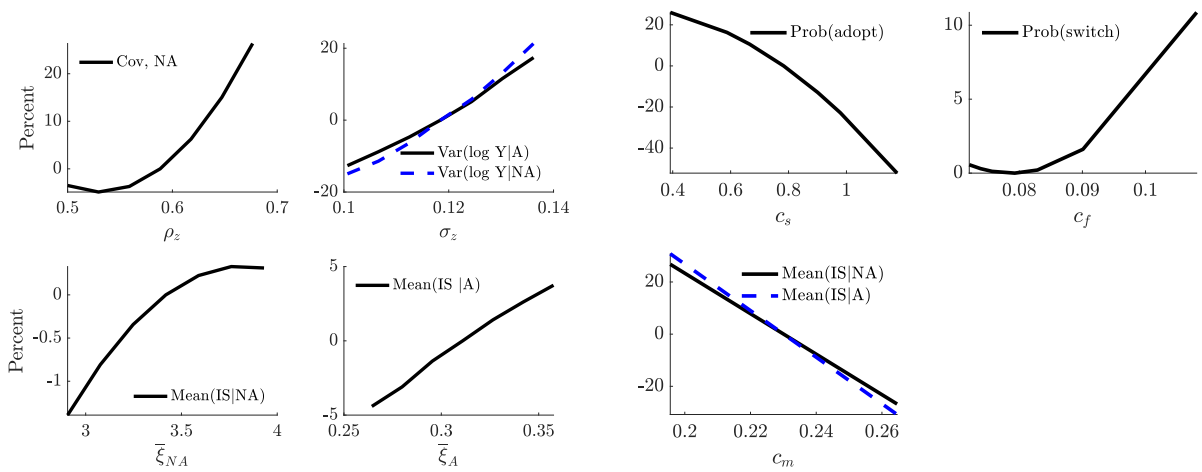
Idiosyncratic productivity persistence mostly informs the covariance between log inventory and log sales. Moreover, the dispersion of idiosyncratic productivity shocks mostly affects variances.

The fixed order costs are strongly related to the mean inventory-to-sales ratios. An increase in the upper bound of fixed order costs for non-JIT adopters leads to an increase in inventory-to-sales for non-JIT producers as these firms stock more inventories in order to incur the higher fixed cost less frequently. Similarly, an increase in the fixed order cost upper bound for JIT firms leads to an increase in the inventory-to-sales ratio for JIT firms.

An increase in the sunk cost of adoption leads to a decrease in the share of JIT adopters. Moreover, an increase in the continuation cost of adoption, causes the share of firms switching out of JIT to rise. Finally, the storage cost also affects the distribution of inventory-to-sales ratios. A higher storage cost raises the marginal cost of carrying inventories across time, and therefore reduces average inventory-to-sales ratios across JIT and non-JIT firms alike.

The ten moments jointly determine the seven parameters that reside in vector θ. Figure C1 reports the monotone relationships between selected moments and parameters. Figure C2 reports the sensitivity of each of the seven parameters to changes in each of the moments. These results

Figure C1: Monotonic Relationships



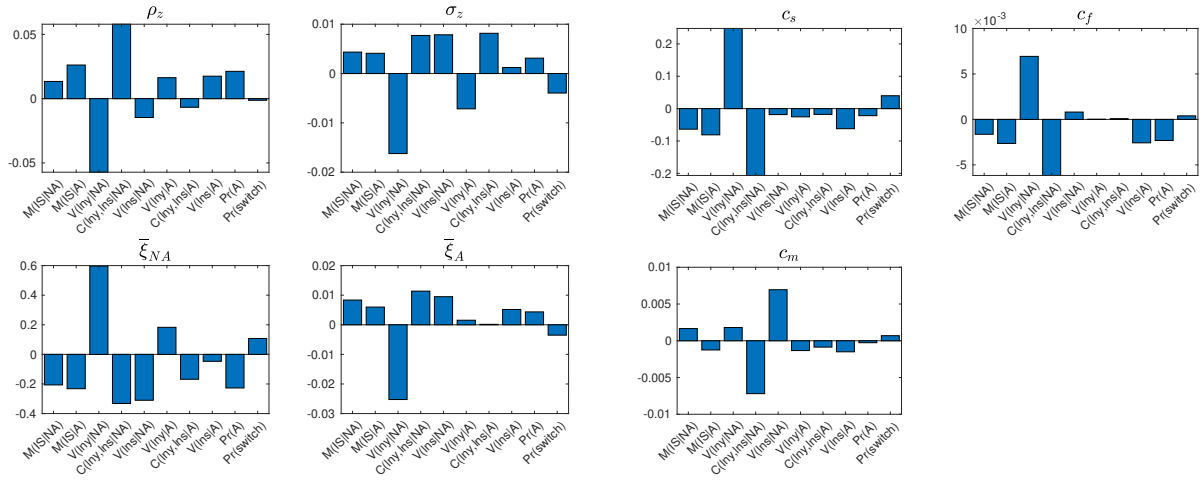
Note: The figure plots the changes in selected moments to changes in the parameters, in percent relative to the moment at the estimated parameter values.

come from an implementation of Andrews et al. (2017). The sensitivity of $\widehat{\theta}$ to $m(\theta)$ is

$$\Lambda = -\left[\left(\frac{\partial m(\theta)}{\partial \theta}\right)' W \left(\frac{\partial m(\theta)}{\partial \theta}\right)\right]^{-1} \left(\frac{\partial m(\theta)}{\partial \theta}\right)' W.$$

I then transform this matrix so that the coefficients reflect the effect on each parameter of a one standard deviation change in the respective moments.

Figure C2: Sensitivity



Note: The figure plots sensitivity estimates as in Andrews et al. (2017). These estimates describe the changes in each of the seven parameters to a one standard deviation increase in each moment.

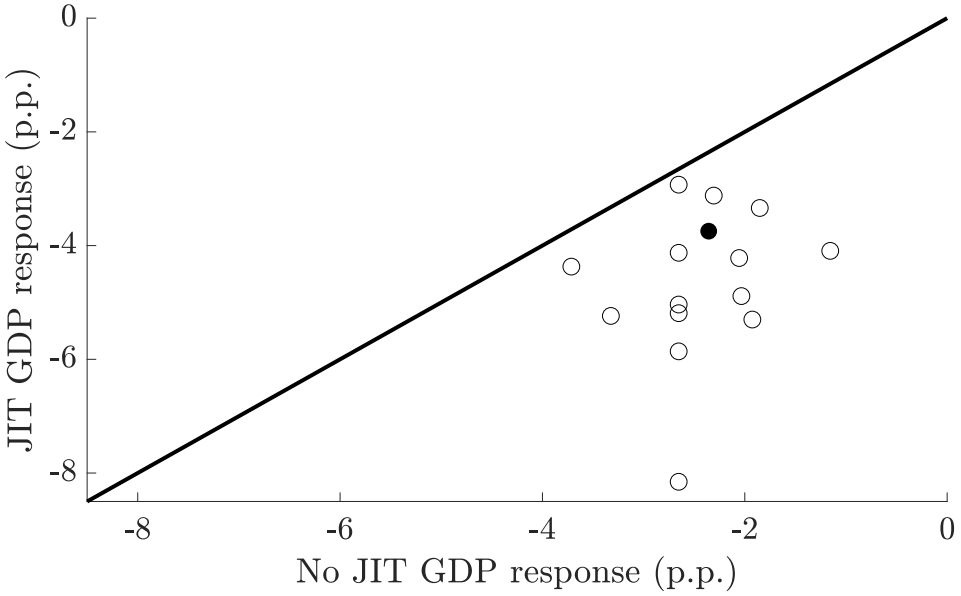
Appendix D Robustness

In this section, I explore the sensitivity of the baseline economy to a fixed order cost shock based on alternative parameterizations, shock sizes, assumptions on the fixed order cost distribution, and under aggregate uncertainty.

D.1 Sensitivity analysis

I start by analyzing the robustness of the unanticipated shock exercise in the main text to different parameterizations. I separately vary each estimated parameter by 5% in either direction, solve for the steady state of the JIT and no JIT economies, and solve for the transition path amid an unanticipated increase in fixed order costs. Figure D1 plots the total output contraction along the transition in each economy. Across all specifications, output contracts more sharply in the JIT economy than in the counterfactual, as shown by the dots residing below the 45-degree line.

Figure D1: JIT Vulnerabilities with Alternative Parameters

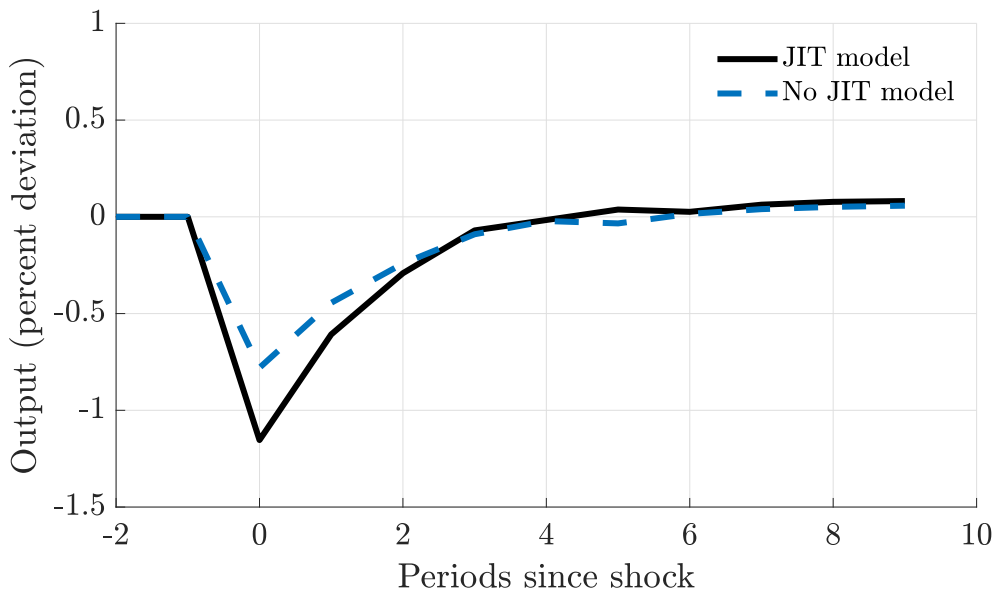


Note: The figure plots the GDP contraction in response to an unanticipated fixed order cost shock in the baseline economy (vertical axis) against the GDP contraction in the counterfactual model (horizontal axis) for a variety of different parameter specifications. The black dot denotes the baseline parameterization estimated in the main text, and the hollow dots represent the alternative parameterizations which reflect 5% deviations in each estimated parameter value in either direction.

D.2 Shock size

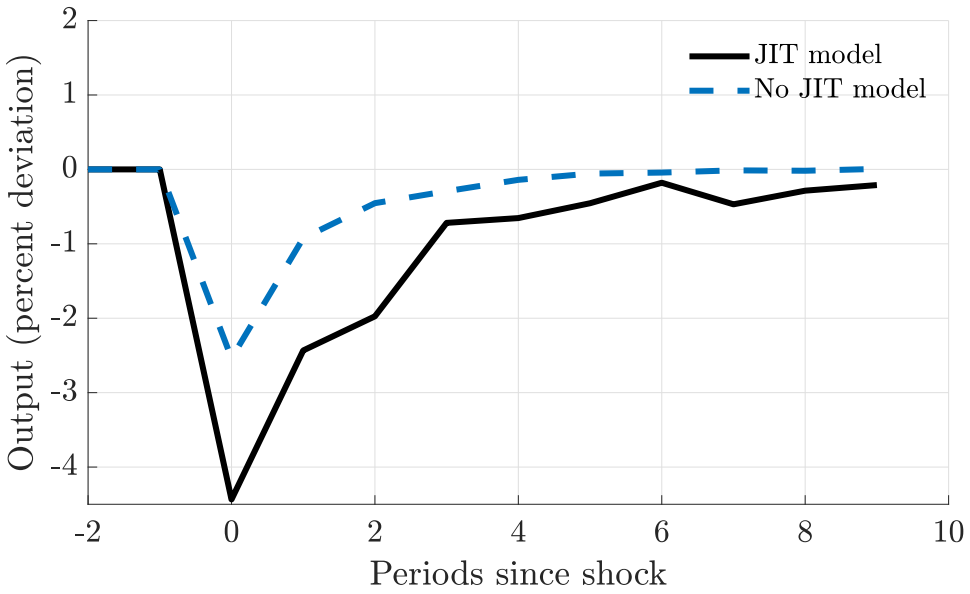
In this section, I consider alternative shock sizes for the unanticipated increase in fixed ordering costs. Figure D2 plots the output dynamics when the shock is one half of the size specified in the main text. Figure D3 plots the output dynamics when the shock is twice the size specified in the main text.

Figure D2: Output Response to Smaller Fixed Order Cost Shock



Note: The figure plots the output response to a fixed order cost shock that is half the size of the shock in Figure 4 of the text. The persistence of the shock is set to 0.50.

Figure D3: Output Response to Larger Fixed Order Cost Shock



Note: The figure plots the output response to a fixed order cost shock that is twice the size of the shock in Figure 4 of the text. The persistence of the shock is set to 0.50.

D.3 Alternative order cost distribution

In this section, I repeat my analysis under an alternative fixed ordering cost distribution. I assume that fixed order costs follow a four-parameter Beta distribution, $\mathcal{B}(\alpha, \beta, \underline{\xi}, \overline{\xi})$, where α and β are the shape parameters of the Beta distribution, and $\underline{\xi}$ and $\overline{\xi}$ are the lower and upper support. The probability density function is:

$$f(\xi) = \frac{(\xi - \underline{\xi})^{\alpha-1}(\overline{\xi} - \xi)^{\beta-1}}{(\overline{\xi} - \underline{\xi})^{\alpha+\beta-1}B(\alpha, \beta)},$$

where $B(\alpha, \beta)$ is the Beta function, $B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt$.

I consider a version of the model with a Beta(3,3,$\underline{\xi}, \overline{\xi}$) order cost distribution, meaning that order costs are symmetrically distributed. Note that the support, $[\underline{\xi}, \overline{\xi}]$ differs for JIT and non-JIT firms, as in the uniform order cost case. Furthermore, as in the case of uniform order costs, I fix $\underline{\xi} = 0$ in the steady state.

First, I find that the economy-wide equilibrium stock of inventories is lower under Beta(3,3) distributed order costs, consistent with the notion that uniformly distributed order costs generate a relatively stronger precautionary inventory holding motive. The first row of Table D1 reports the aggregate inventory stock in Beta order cost model relative to the baseline model. The second row reports the relative output in the JIT model compared to the counterfactual model prior to the shock.

Table D1: Aggregate Outcomes Under Beta-Distributed Order Cost

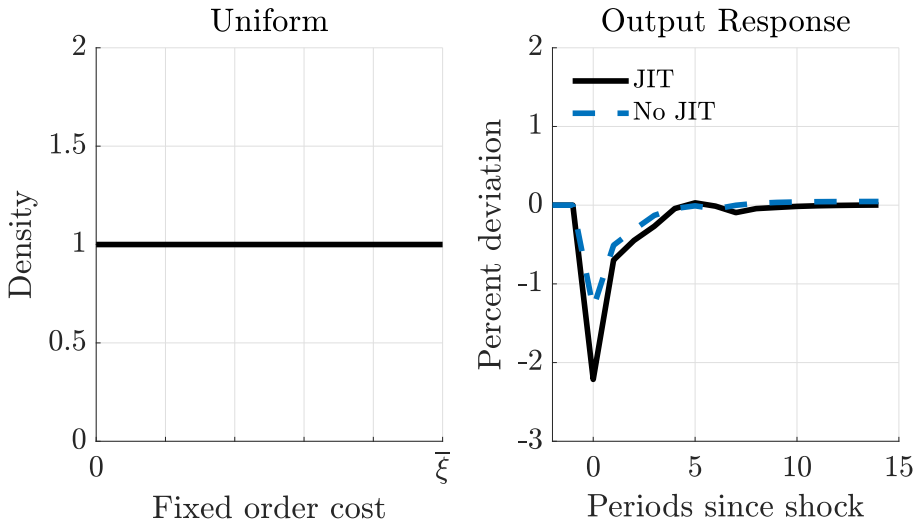
	Uniform	Beta(3,3)
Steady state inventory stock relative to uniform (%)	–	-23.88
Output relative to no JIT counterfactual before shock (%)	15.28	22.17

Note: The first row of the table reports the steady state inventory stock relative to the baseline assumption of uniformly distributed fixed order costs. The second row of the table reports the output prior to the shock in the JIT economy relative to output prior to the shock in the analogous no-JIT economy.

Figures D4 and D5 plot the transition dynamics under uniform and Beta(3,3) distributed fixed

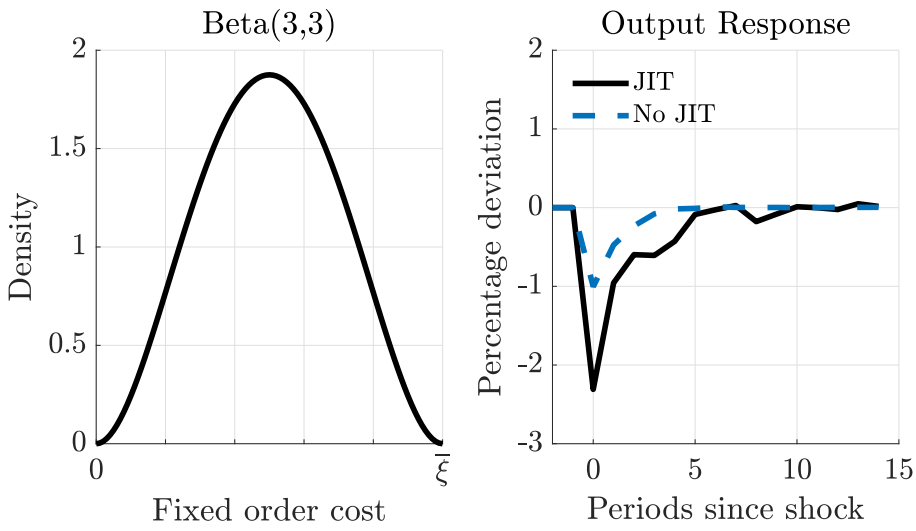
order costs. Overall, the results are quite similar across order cost specifications. The JIT economy under Beta(3,3) experiences a slightly larger output contraction amid a shock to fixed order costs relative to the version with uniformly distributed fixed order costs.

Figure D4: Output Response with Uniformly Distributed Order Costs



Note: The left panel plots the assumed pdf of fixed ordering cost in the model. The right panel plots the output response to a fixed order cost shock, as in Figure 4 of the main text.

Figure D5: Output Response with Beta(3,3) Distributed Order Costs



Note: The left panel plots the assumed pdf of fixed ordering costs in the model. The right panel plots the output response to a fixed order cost shock. The shock is equal in size and persistence to the one used in Figure 4 of the main text.

D.4 Incorporating aggregate uncertainty

The supply disruption that I model in the main text takes the form of an unanticipated fixed order cost shock. After the realization of the shock, agents have perfect foresight about the transition back to steady state. In this section, I show that my findings generally extend to an environment in which agents form expectations over an aggregate shock to fixed order costs.

D.4.1 Aggregate uncertainty about fixed order costs

I assume that the support of fixed order costs is time varying,

$$\underline{\xi}_t = \underline{\xi} + x_t \quad \text{and} \quad \overline{\xi}_t = \overline{\xi} + x_t.$$

In addition, I assume that x_t is a two-dimensional state, $x_t = [0.0 \;\; 0.054]$, where $x_t = 0.0$ reflects normal times and $x_t = 0.054$ reflects a supply disruption, which is calibrated to be the same as in the unanticipated shock exercise. I set the transition matrix to be,

$$\Pi(x'|x) = \begin{bmatrix} 0.95 & 0.05 \\ 0.25 & 0.75 \end{bmatrix},$$

which implies that supply disruptions are infrequent and that a transition from a supply disruption back to normal times is relatively quick. The aggregate state space is now comprised of x and μ, the distribution of firms, and we can denote it as $\Psi = (x, \mu)$.

Solving the JIT model with aggregate shocks requires tracking prices and the distribution of firms, an infinite-dimensional object, across time. Following Krusell and Smith (1998), I solve the model by assuming that agents exhibit bounded rationality and use aggregate material input usage to track the distribution across time. I define a log linear mapping between prices and aggregate

materials,

$$\log \widehat{M}'(x) = \beta_0^M(x) + \beta_1^M(x) \log M(x)$$
$$\log \widehat{p}(x) = \beta_0^p(x) + \beta_1^p(x) \log M(x).$$

Below I summarize the steps taken to solve the model:

1. Simulate $\{X_t\}_{t=1}^T$ for some large T and guess an initial set of coefficients: $\beta_0^{M(0)}$, $\beta_1^{M(0)}$, $\beta_0^{p(0)}$, and $\beta_1^{p(0)}$.

2. For each iteration i:

   (a) Solve the final goods firm problem on a grid based on $\widehat{p}^{(i)}$ and $\widehat{M}'^{(i)}$, where $\widehat{p}^{(i)}$ and $\widehat{M}'^{(i)}$ are obtained using the coefficients $\beta_0^{M(i)}, \beta_1^{M(i)}, \beta_0^{p(i)}, \beta_1^{p(i)}$.

   (b) Simulate the model, using $\{X_t\}_{t=1}^T$, and obtain a time series $\{p_t, M_t\}_{t=1}^T$, where p_t reflects the equilibrium price which is obtained by clearing markets in each period.

   (c) Based on the simulated data, update the forecast rules via OLS to obtain $\beta_0^{M(i+1)}$, $\beta_1^{M(i+1)}$, $\beta_0^{p(i+1)}$, and $\beta_1^{p(i+1)}$. If the coefficients are sufficiently close, up to a pre-specified tolerance, then exit. Otherwise, update each coefficient as a convex combination of the old guess and the new estimate. Set $i = i + 1$ and return to (a).

I specify a grid of dimension $N_z \times N_s \times N_a \times N_X \times N_M = 5 \times 120 \times 2 \times 2 \times 10$, and I simulate the model for 5,500 periods, discarding the first 500 periods to reduce the influence of initial conditions. I specify a tolerance of 10^{-3} for convergence of the forecasting rules.

D.4.2 Solution and accuracy statistics

The converged forecast rules are reported in Table D2. Table D3 reports accuracy statistics for each model based on static and dynamic forecasts for aggregate materials and prices. The first row of the table reports the mean percentage difference between realized simulated data and its corresponding

dynamic forecast based on the forecasting rules (Den Haan, 2010). The bottom two rows report the R^2 of each forecast rule based on static forecasts.

Table D2: Forecasting Rules

	Baseline	Counterfactual
$\beta_0^M(x = x_1)$	-1.536	-2.759
$\beta_0^M(x = x_2)$	-1.806	-2.695
$\beta_1^M(x = x_1)$	0.397	-0.041
$\beta_1^M(x = x_2)$	0.297	-0.010
$\beta_0^p(x = x_1)$	1.219	2.165
$\beta_0^p(x = x_2)$	1.578	2.210
$\beta_1^p(x = x_1)$	-0.373	-0.044
$\beta_1^p(x = x_2)$	-0.240	-0.031

Note: The table reports the forecasting rules.

Table D3: Accuracy Statistics

	Baseline		Counterfactual	
	M	p	M	p
Mean percentage difference	0.633	0.378	0.437	0.009
Forecast regression R^2				
$x = x_1$	1.000	1.000	1.000	1.000
$x = x_2$	1.000	1.000	1.000	1.000

Note: The table reports accuracy statistics.

D.4.3 Impulse response to a fixed order cost shock

To compute the impulse response to a shock to x_t, I follow the Koop et al. (1996) approach and compute the generalized impulse response. More specifically, I simulate 5,000 economies for 50 periods. I do this twice: once for a set of simulations that does not impose the shock, and again for simulations which do impose the shock. I then compute the impulse response for GDP as,[3]

$$\text{GDP}_t^{\text{IRF}} = \frac{1}{5000} \sum_{j=1}^{5000} \left[\log \left(\frac{\text{GDP}_{jt}^{\text{shock}}}{\text{GDP}_{jt}^{\text{no shock}}} \right) \times 100 \right].$$

Figure D6 plots the impulse response in the baseline and counterfactual economies. The top left panel shows that output initially increases in the baseline economy relative to the counterfactual economy, but subsequently contracts further and recovers more gradually. The top right panel plots sales, which contracts more in the baseline model relative to the counterfactual without an initial relative increase.

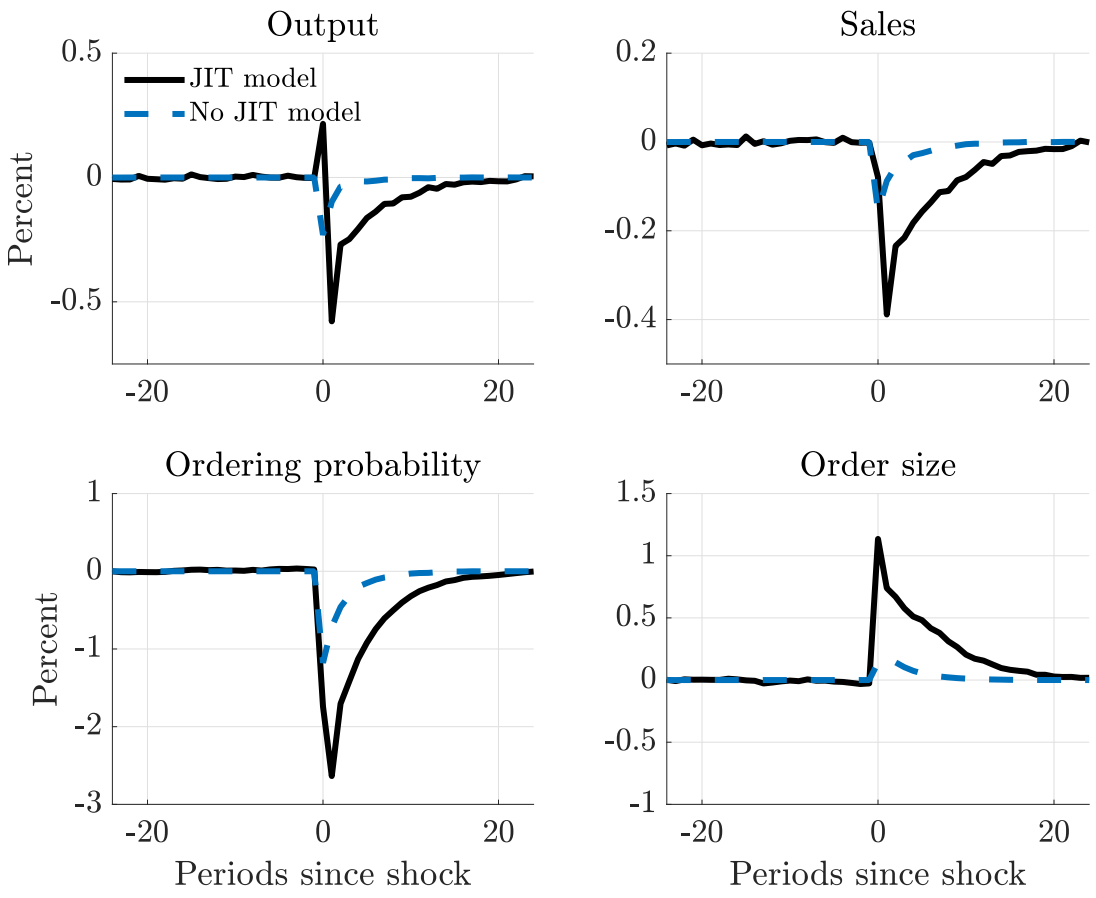
The relative increase in output on impact is therefore due to a relative increase in inventory investment on impact which is ultimately attributed to the behavior of aggregate orders in the model. Under aggregate uncertainty, firms recognize the likelihood of transiting to the high fixed order cost state and optimally hold precautionary stocks of inventory in response. This, in turn, renders ordering probabilities less sensitive on impact to a fixed order cost shock.[4] The aggregate ordering probability declines by about 1.7% on impact but then falls further as inventories are depleted, reaching a trough of -2.6% one period after the shock. Meanwhile, order sizes increase on impact. Taken together, this yields a relative increase in aggregate orders in the period in which the aggregate shock is realized.

After the period in which the shock occurs, however, output declines by more in the baseline economy than in the counterfactual economy as inventory buffers are rundown and aggregate orders decline. Therefore, the findings documented in this section are broadly consistent those obtained

[3] See Koop et al. (1996) for more details. Terry (2017) also provides a detailed summary of this approach.

[4] One way to see this by noting that the slope of ordering probabilities in panel (a) of Figure 3 is flatter at higher levels of inventory. Though ordering probabilities still decline more in the baseline economy, they do so less than in the unanticipated shock exercise.

Figure D6: Output Response to Fixed Order Cost Shock



Note: The figure depicts the response of output to a positive shock to fixed order costs, averaged over 5,000 simulated economies.

from the unanticipated shock exercise and indicate that the vulnerability of JIT producers to supply disruptions is also present in an environment that features aggregate uncertainty, with the caveat that the extent of this vulnerability is shaped in part by the size of the shock and the degree to which firms hold precautionary stocks of inventories.[5]

[5]The initial increase in aggregate orders may be tempered, or potentially even disappear, in an environment with sufficiently large capital adjustment costs, as capital is an input in the production of orders and such a friction would dampen the investment response to the shock. I abstract away from capital adjustment costs in my model for simplicity.

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